Advanced Neural Data Analysis (ANDA) 2026

Cortical Variability Dynamics - Exercises

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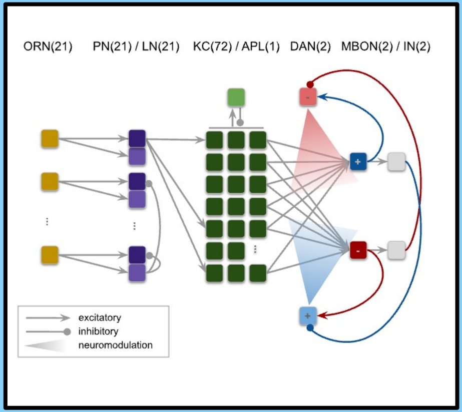
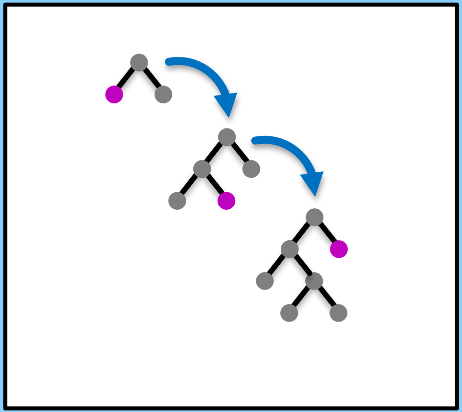
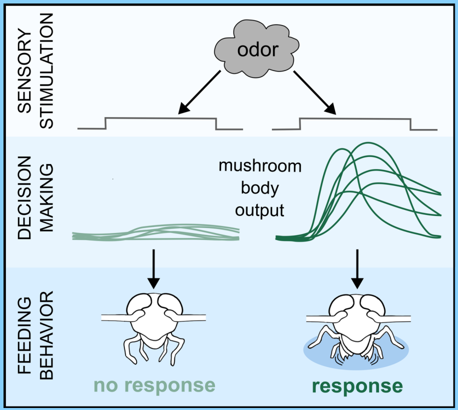
Computational Models of Nervous System Function



Artifical Intelligence for Neuroscience



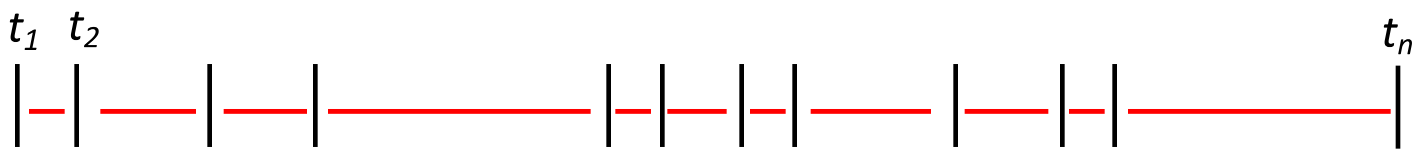
Behavioral Neurophysiology



Spike Train Statistics

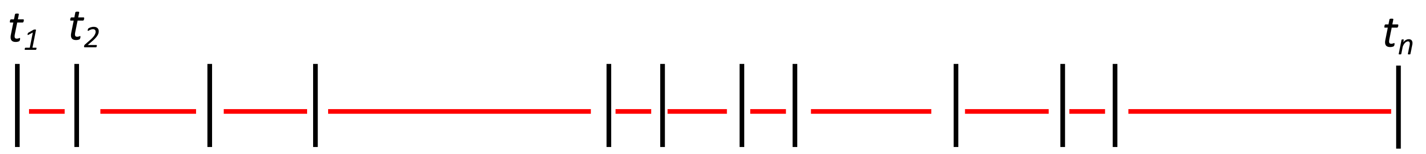
concept & theory

Interval and Count: Random Variables

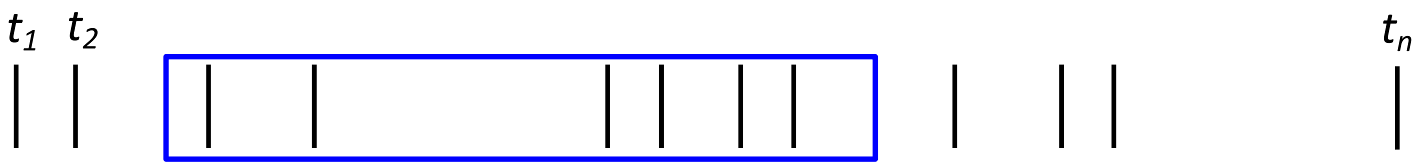


X_i = *Inter Spike Interval (ISI), continuous random variable*

Interval and Count: Random Variables

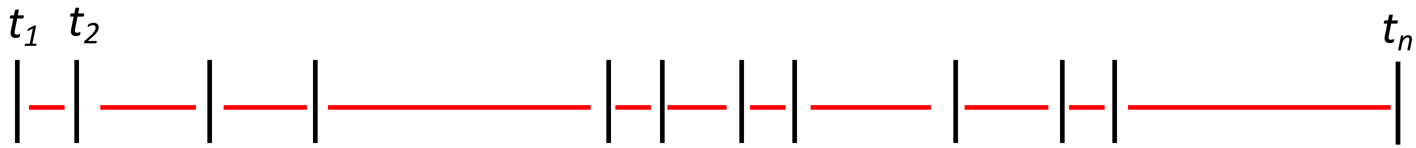


$\boldsymbol{X_i}$ = *Inter Spike Interval (ISI), continuous random variable*

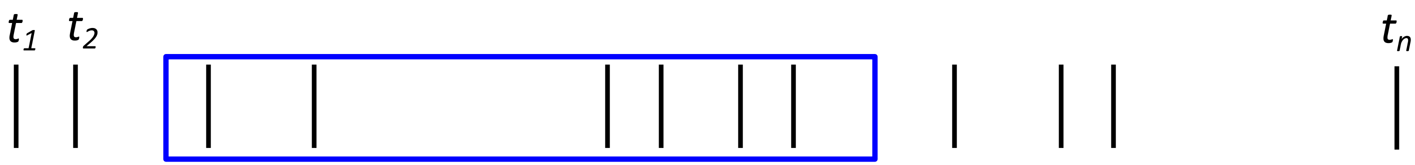


$\boldsymbol{N_i}$ = *Spike Count, discrete random variable*

Interval and Count: Random Variables



X_i = *Inter Spike Interval (ISI), continuous random variable*



N_i = *Spike Count, discrete random variable*

N_i *and* X_i *are related for any given point process:*

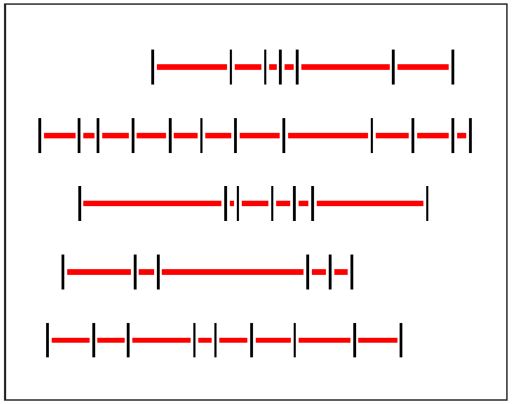
$$P\{\tau_k \leq t\} = P\{N_{[0,t)} \geq k\} \qquad \tau_k = \sum_{i=1}^{k} X_i$$

Assumptions : stationary and orderly point process

Interval and Count Variability: Expectation

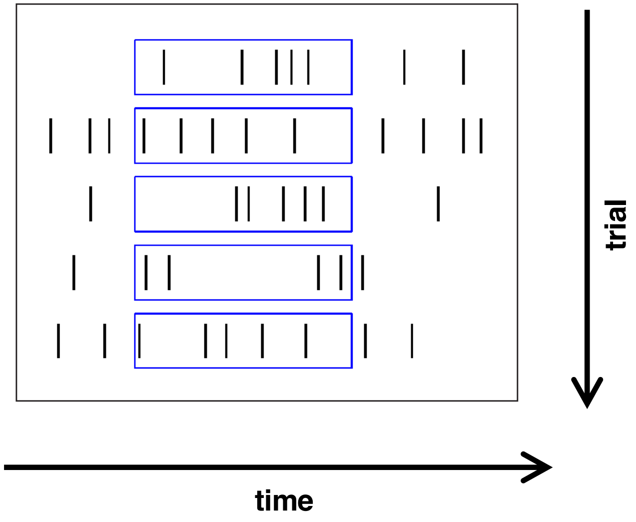
Coefficient of variation
(of inter-spike intervals *X*)

$$CV^2 = \frac{Var(ISI)}{mean^2(ISI)}$$



Fano factor
(of spike count N)

$$FF = \frac{Var(count)}{mean(count)}$$



Interval and Count Variability: Expectation

Coefficient of variation
(of inter-spike intervals *X*)

$$CV^2 = \frac{Var(ISI)}{mean^2(ISI)}$$

Fano factor
(of spike count N)

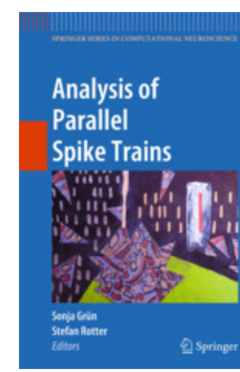
$$FF = \frac{Var(count)}{mean(count)}$$

$$\boxed{\lim_{T \to \infty} \mathrm{FF} = \mathrm{CV}^2_\infty (1 + 2\,\xi) \quad \text{with } \xi = \sum_{i=1}^{\infty} \xi_i}$$

where ξ_i denote coefficients of serial interval correlation.

Thus, for equilibrium **renewal processes** we expect:

$$\boxed{\mathbf{FF \approx CV^2}}$$

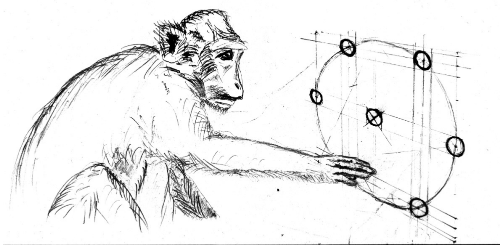


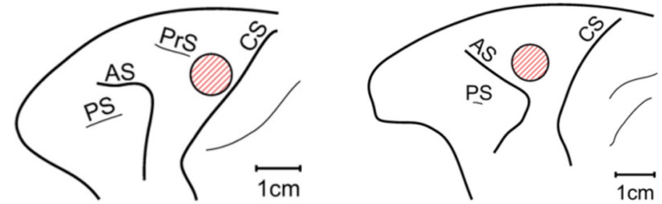
Cox and Lewis (1966); Farkhooi, Müller, Nawrot (2011) Phys Rev E 83; Methods: Nawrot (2010)

Spike Train Statistics

in the behaving monkey

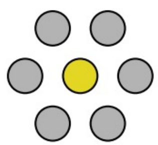
Delayed Reach-Task with Varying Uncertainties

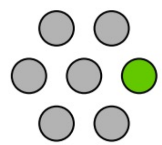




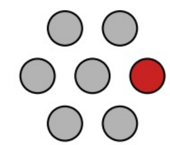
TS | PS | | RS |
---|---|---|---|---

C1 certainty = 1





Delay Period

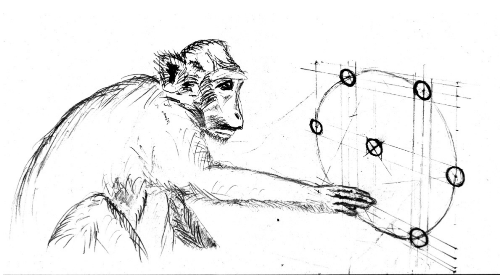


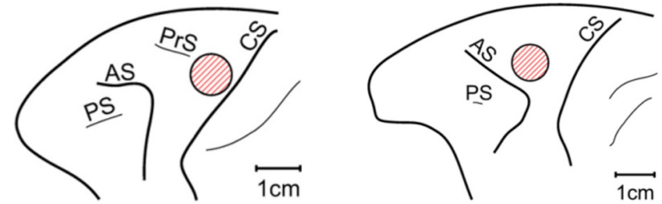
Movemen
t

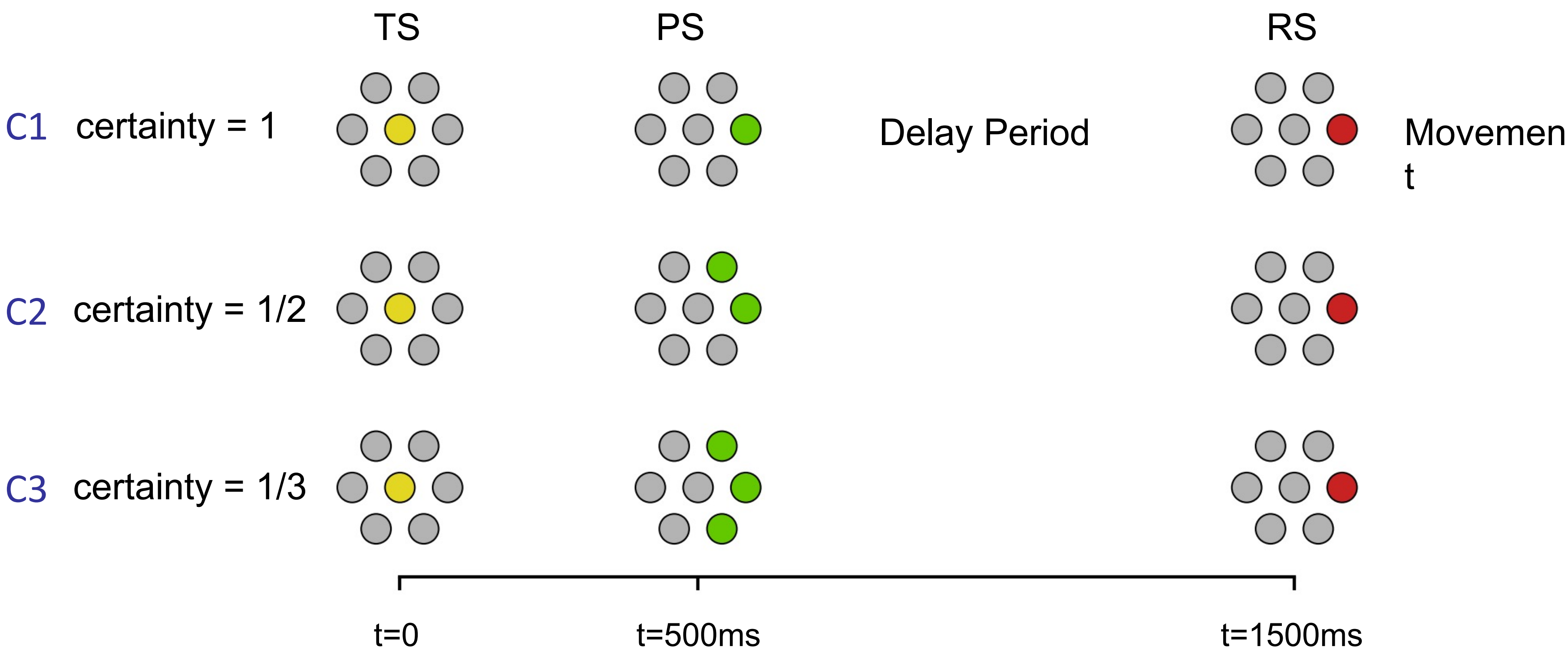
t=0 t=500ms t=1500ms

Rickert et al. (2009) J. Neuroscience
Rostami et al. (2024) Nature Comm

Delayed Reach-Task with Varying Uncertainties







Rickert et al. (2009) J. Neuroscience
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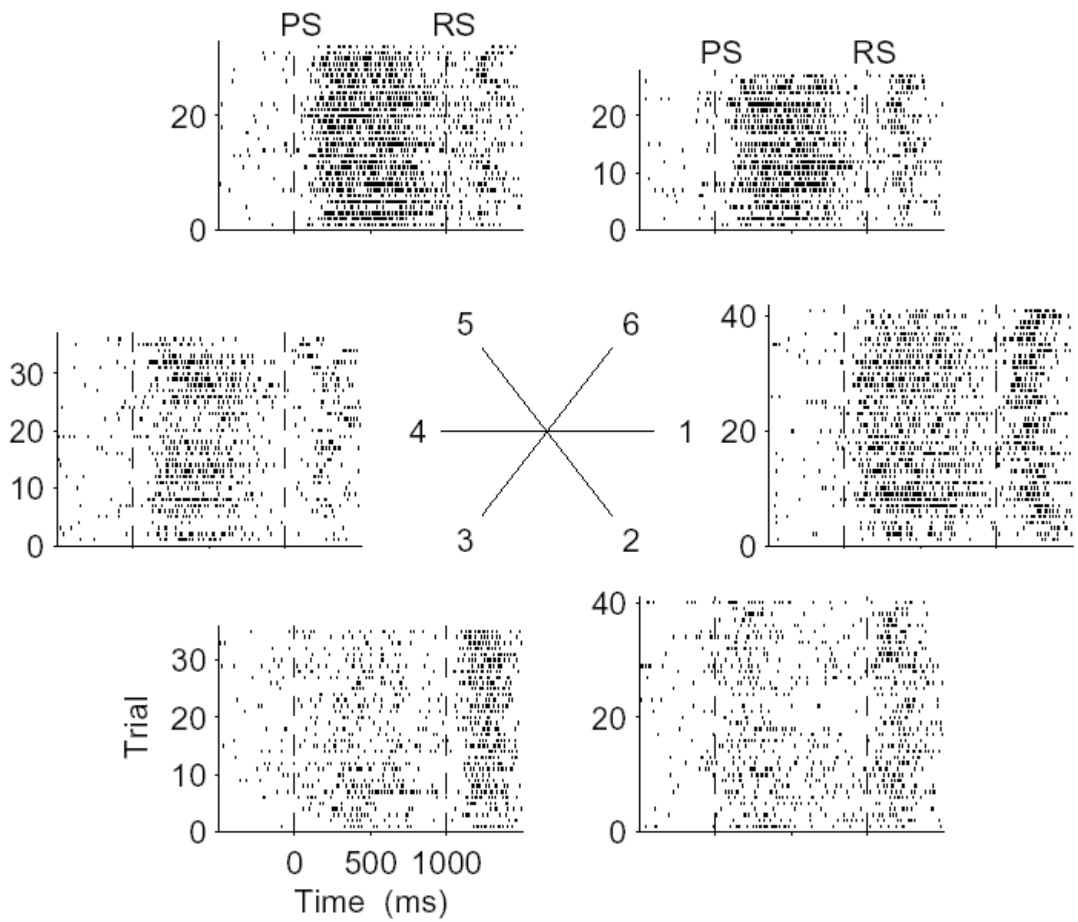
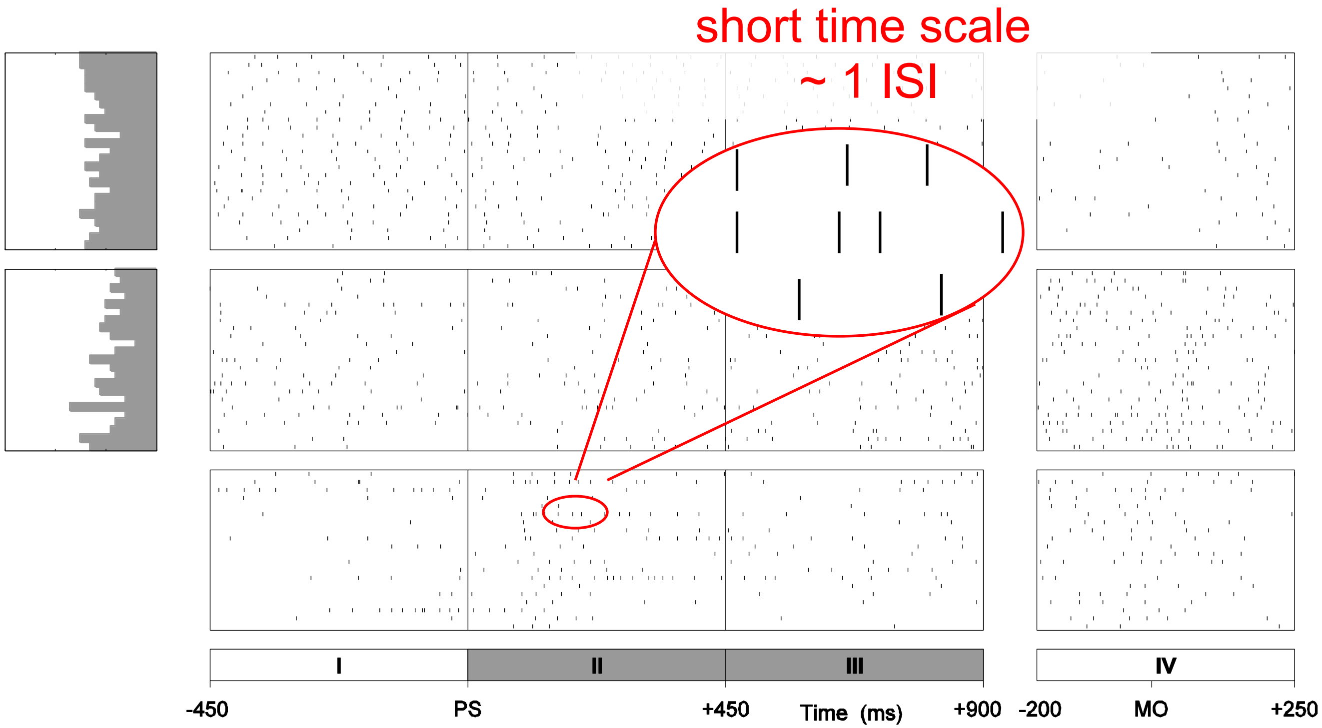
Single Unit Activity During Reach-Task



Rickert et al. (2009) J. Neuroscience

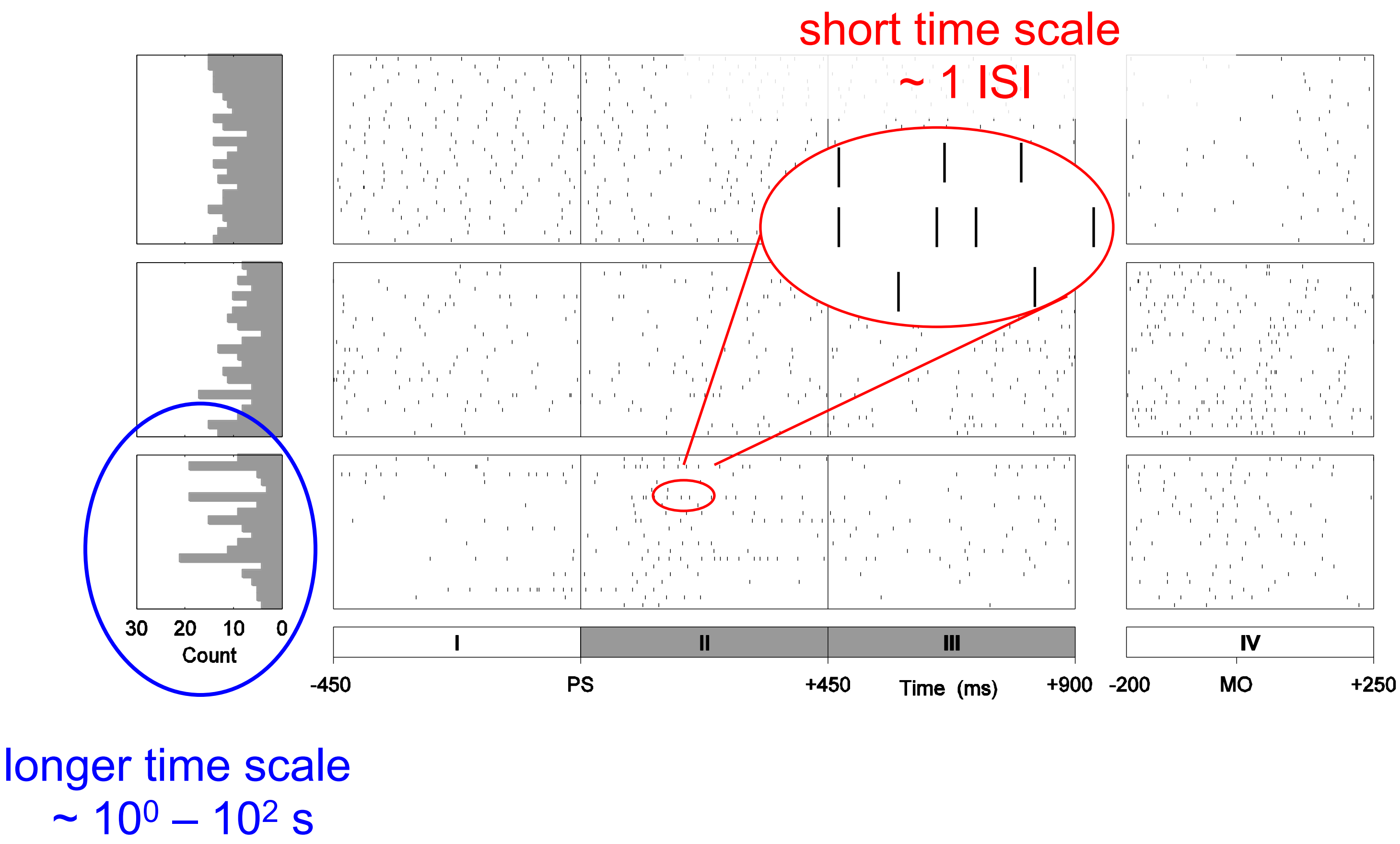
Task-related Variability in the Motor Cortex

- Interval variability: CV / CV_2 / L_v

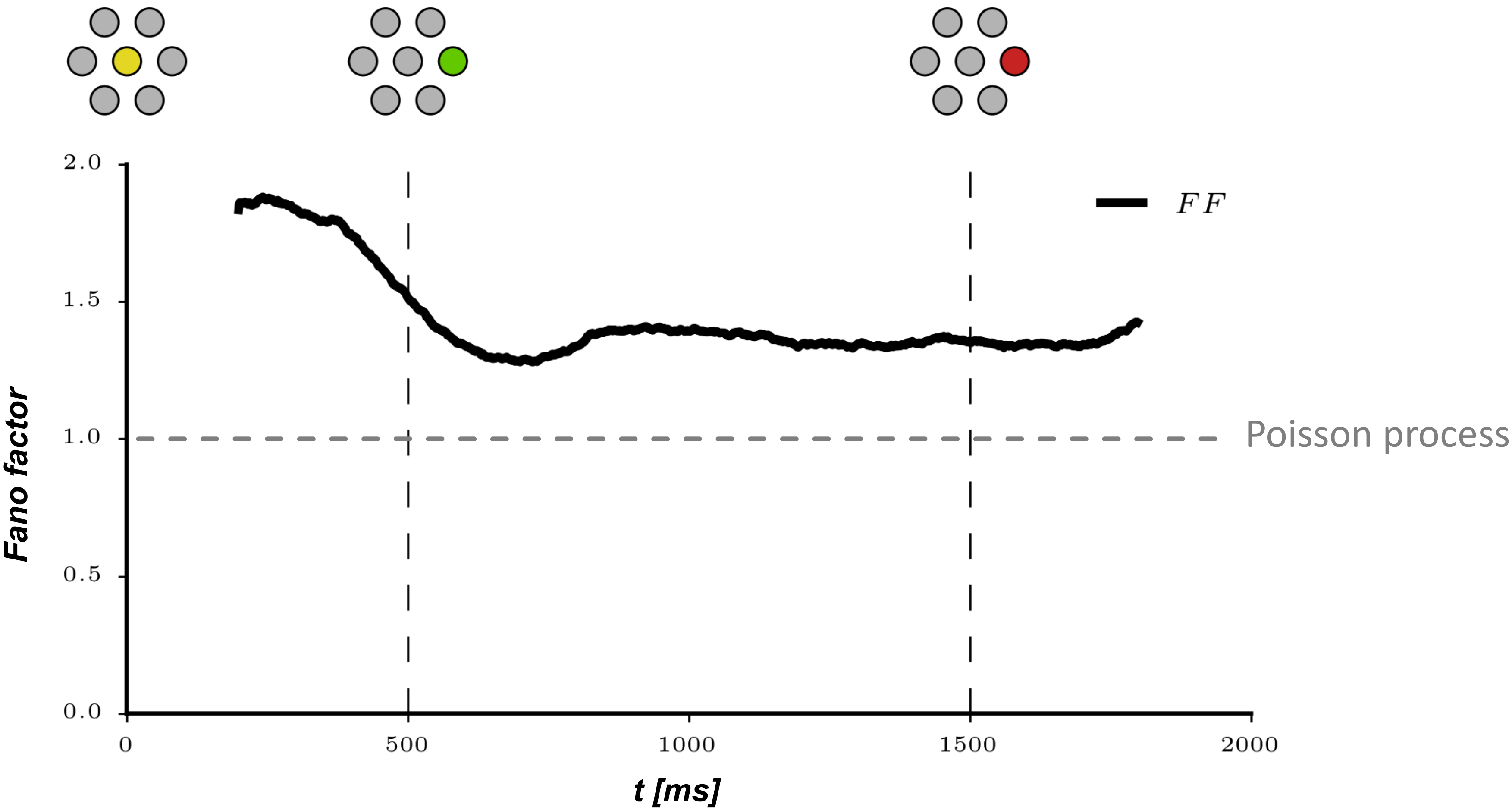


Task-related Variability in the Motor Cortex

- Interval variability: CV / CV_2 / L_v
- Spike count variability: FF



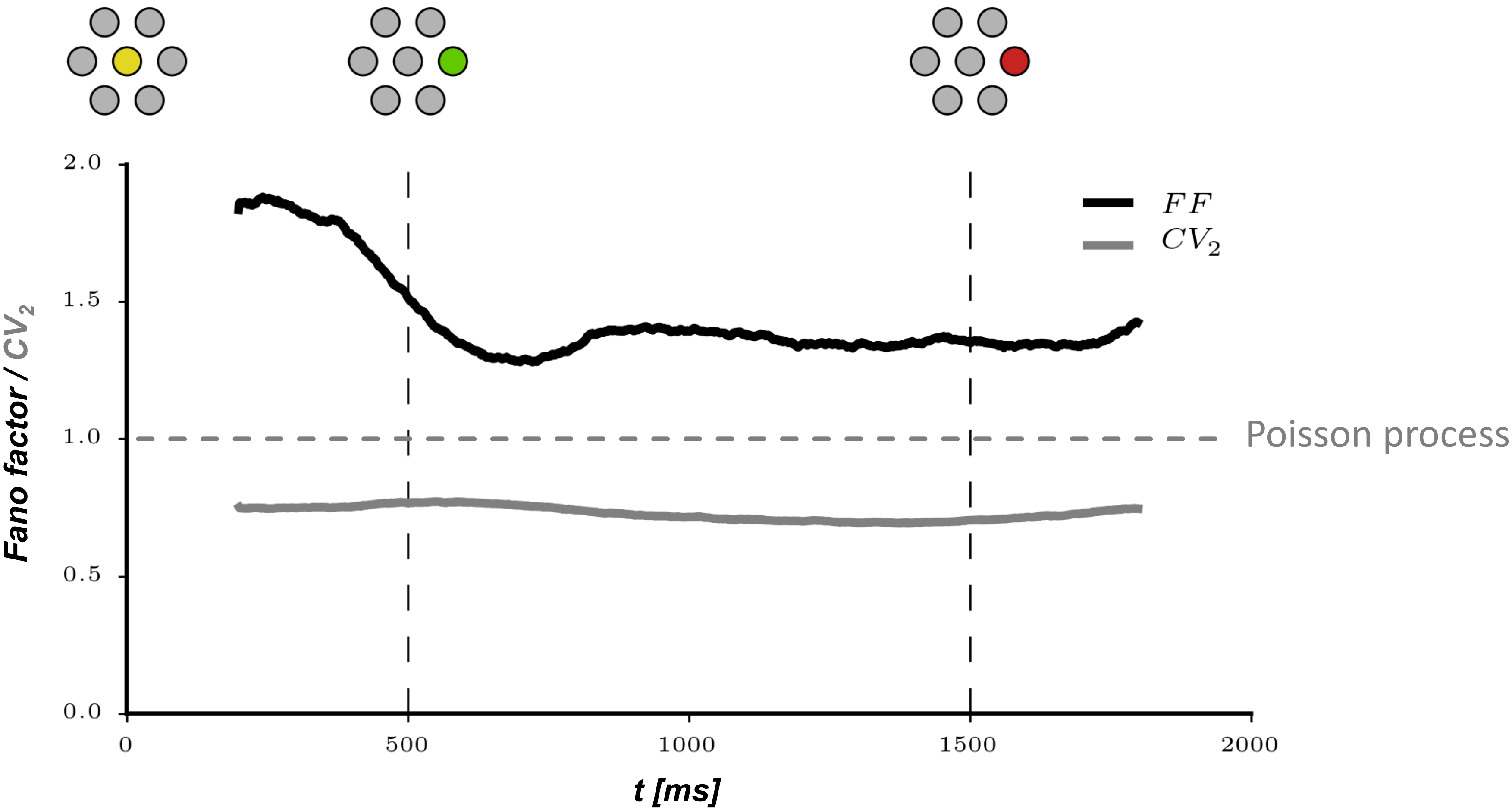
Task-related Variability Dynamics *in vivo*



Rostami et al. (2024) Nat Comm
Rickert et al. (2009) J Neurosci
Nawrot (2003) PhD Thesis

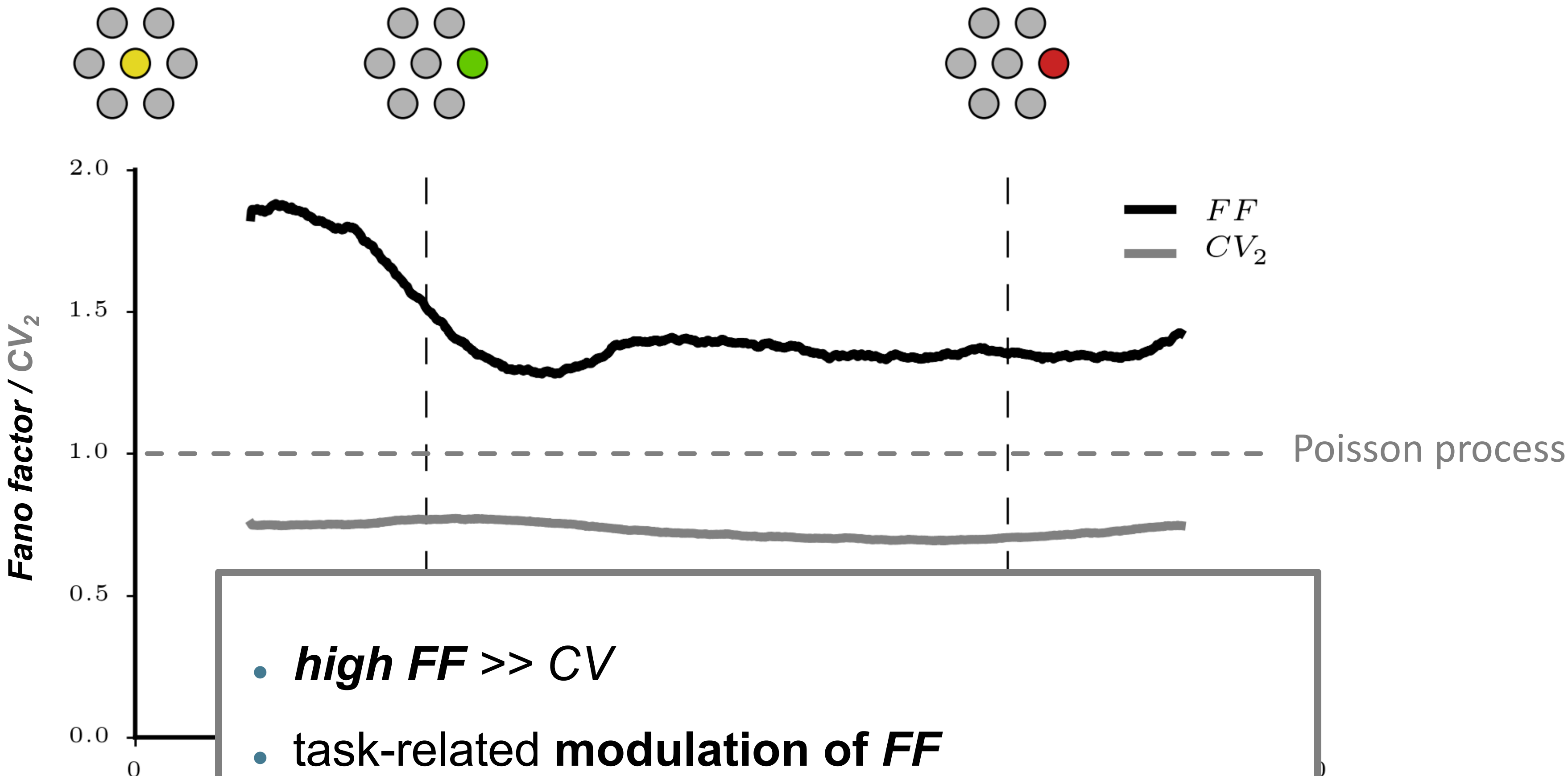
Churchland et al. (2010) Nat Neurosci
Churchland et al. (2006) J Neurosci

Variability Dynamics *in vivo*



Rostami et al. (2024) Nat Comm
Riehle et al. (2018) Front Neural Circuits

Variability Dynamics *in vivo*



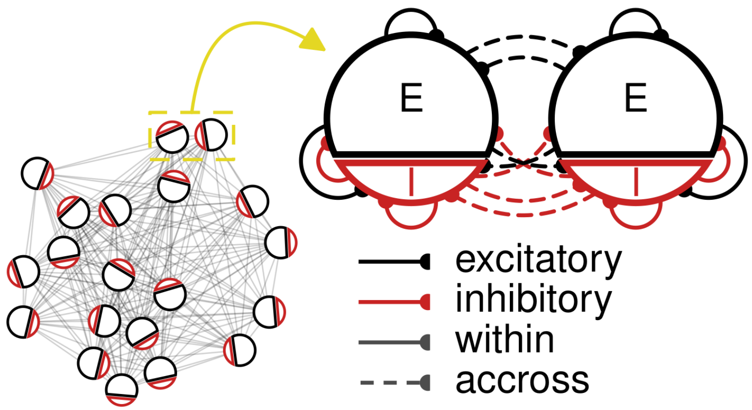
- ***high FF*** >> *CV*
- task-related **modulation of *FF***
- ***CV*** is **constant**

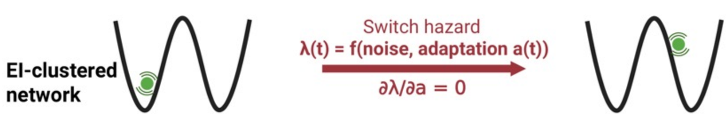
➔ ***Which mechanisms rule this dynamics ?***

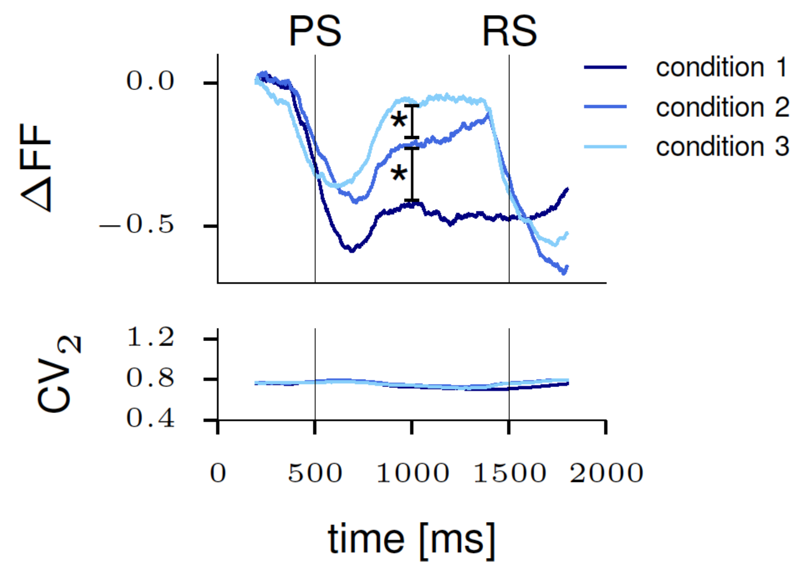
Mechanistic Model

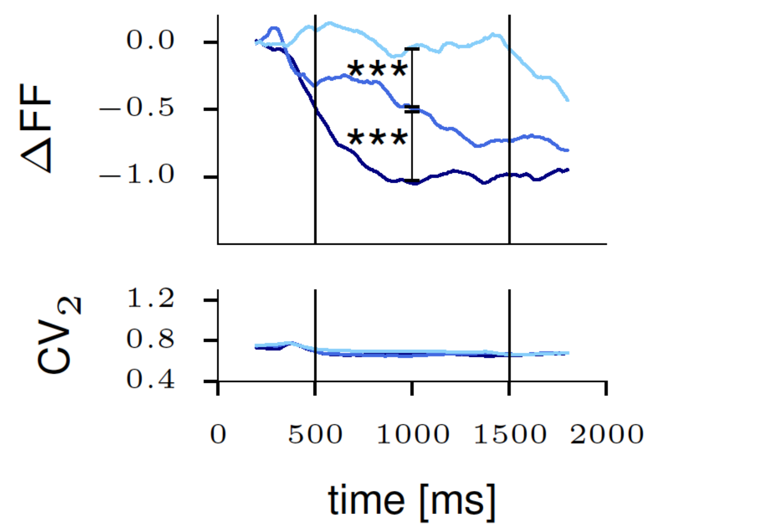
for cortical count and interval variability

Point-Attractor Models support high modulated FF and constant CV









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